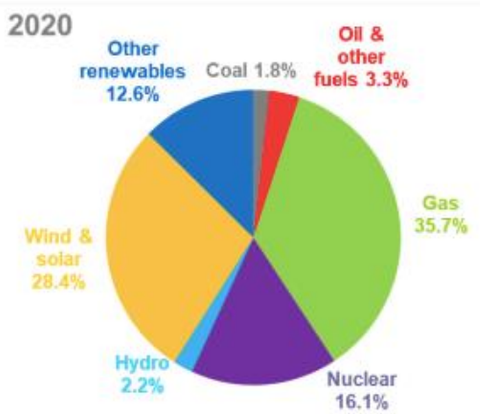
Knowledge Organiser
Year 7: Energy – Energy Resources

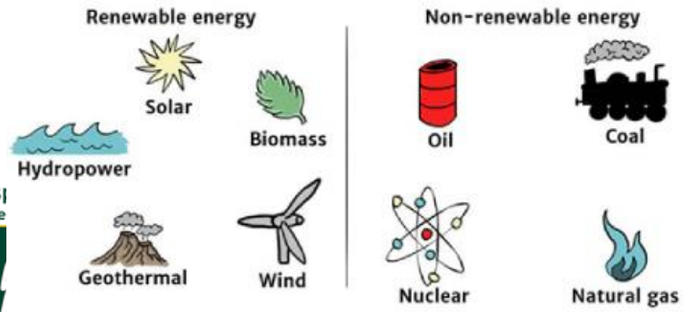
How do we generate our electricity?
- Electricity is generated by transferring energy from another store e.g. the chemical store in natural gas
- The energy released from these stores is used to power a generator, which produces an electrical current

A pie chart showing the approximate **energy mix** in the UK in 2020:



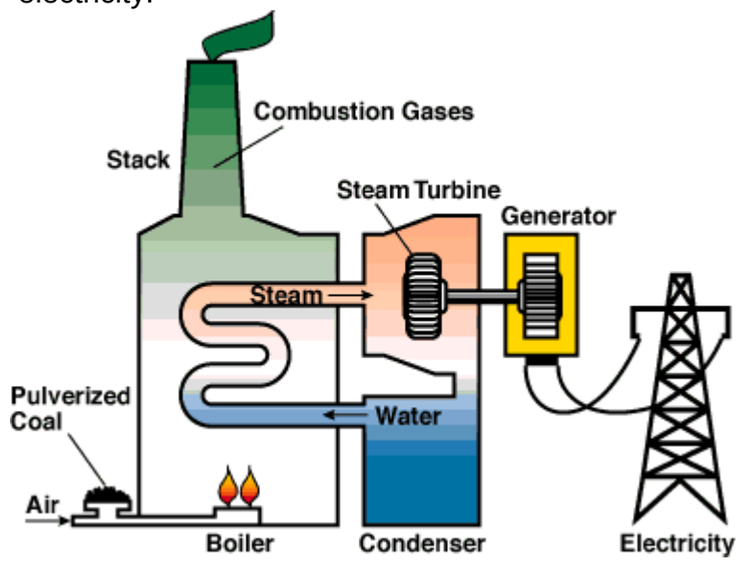
What is a non-renewable source of energy?
- A non-renewable resource is one that is ***used up at a faster rate than it can be replaced***
- Fossil and nuclear fuels are the two most common non-renewable energy resources
- Fossil fuels like coal take millions of years to form, but can be burnt very easily and quickly
- These resources are not sustainable, some day they will run out

A table showing a selection of renewable and non-renewable energy resources:



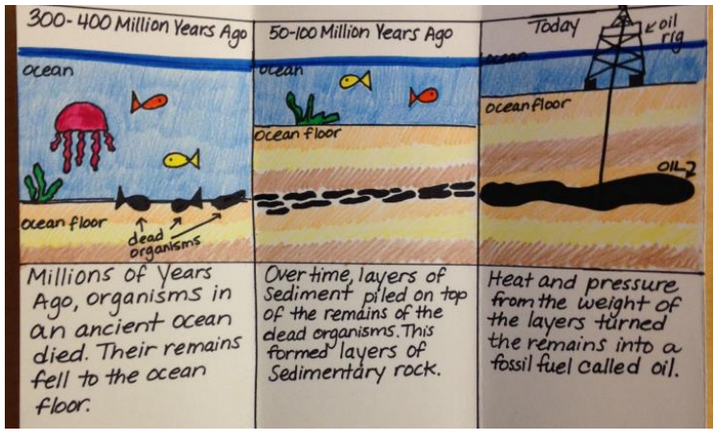
What are fossil fuels?
- Fossil fuels are the remains of long dead plants and animals
- There are three types of fossil fuel; coal, crude oil and natural gas
- The chemical energy stored in their biomass is contained within the fossil fuel
- This can be transferred into the thermal energy store by burning the fuels

A diagram showing how coal can be burnt to generate electricity:



How do fossil fuels generate electricity?
- When the fossil fuel is burnt, chemical energy is transferred into the thermal store of the water
- The water heats up and evaporates into steam
- The steam pushes a turbine round, which runs a generator
- The generator transfer kinetic energy into electrical potential

A diagram showing how fossil fuels are formed:



How are fossil fuels formed?
- Fossil fuels are formed from the fossilised remains of plants and animals
- Over millions of years the organic molecules inside their bodies are broken down
- Immense heat and pressure are required to do this

What are renewable resources?
- Renewable energy resources are ones which ***can be replaced at the same rate they are used up***
- Examples include wind turbines, solar panels, hydroelectric dams and biomass furnaces
- Renewable resources are more sustainable than fossil fuels because they do not run out
- They also have a lower **environmental impact**

A table which compares fossil fuels, nuclear power, bio fuels and wind power

Energy Resource	Renewable?	Advantages	Disadvantages
Fossil fuels	No	Reliable. Can produce large amounts of energy at fairly short notice.	Produces significant greenhouse gases and pollution
Nuclear	No	Reliable. Produces no greenhouse gases or pollution. A large amount of energy is produced from a small amount of fuel.	Produces dangerous radioactive waste that can take thousands of years to decay
Bio-fuels	Yes	The CO_2 produced while burning the fuel is balanced by the CO_2 absorbed whilst producing it.	Can take up a lot of land and consume resources that are needed for food production
Wind	Yes	Produces no greenhouse gases or pollution. Land can still be used for farming.	Not reliable. Turbines can be noisy and ugly. Not everywhere is suitable

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RAG Yourself!

Learning Outcomes;	RAG
State the meaning of non-renewable	
State the three types of fossil fuel	
Explain how fossil fuels generate electricity	
State examples of renewable energy resources	
Explain how solar, wind and tidal work	
Evaluate the use of renewable and fossil fuels	

Knowledge Organiser
Year 7: Energy – Energy Resources

How do we generate our electricity?

A pie chart showing the approximate **energy mix** in the UK in 2020:

What is a non-renewable source of energy?

A table showing a selection of renewable and non-renewable energy resources:

What are fossil fuels?

A diagram showing how coal can be burnt to generate electricity:

How do fossil fuels generate electricity?

A diagram showing how fossil fuels are formed:

How are fossil fuels formed?

What are renewable resources?

A table which compares fossil fuels, nuclear power, bio fuels and wind power

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